Using the same argument for fitting the regression equation of Y on X, we have the simple linear regression equation of X on Y with best fit as

$$\hat{X} = \hat{c} + b_{XY}\, y$$

where $\hat{c} = \bar{x} - b_{XY}\,\bar{y}$

$$b_{XY} = \frac{\frac{1}{n}\sum_{i=1}^{n} x_i y_i - \bar{x}\,\bar{y}}{\frac{1}{n}\sum_{i=1}^{n} y_i^2 - \bar{y}^2}$$

The slope-point form of this equation is

$$\hat{X} - \bar{x} = b_{XY}(y - \bar{y}).$$

Also, the relationship between the Karl Pearson's coefficient of correlation and the regression coefficient are

$$b_{XX} = r_{XY}\frac{SD(X)}{SD(Y)} \text{ and } b_{YX} = r_{XY}\frac{SD(Y)}{SD(X)}.$$

5.5 PROPERTIES OF REGRESSION COEFFICIENTS

1. Correlation coefficient is the geometric mean between the regression coefficients.

$$r_{XY} = \sqrt{b_{XY} \times b_{YX}}$$

2. It is clear from the property 1, both regression coefficients must have the same sign. *i.e.,* either they will positive or negative.
3. If one of the regression coefficients is greater than unity, the other must be less than unity.
4. The correlation coefficient will have the same sign as that of the regression coefficients.
5. Arithmetic mean of the regression coefficients is greater than the correlation coefficient.

$$\frac{b_{XY} + b_{YX}}{2} \geq r_{XY}$$

6. Regression coefficients are independent of the change of origin but not of scale.

Properties of regression equation

1. If $r = 0$, the variables are uncorrelated, the lines of regression become perpendicular to each other.
2. If $r = 1$, the two lines of regression either coincide or parallel to each other.
3. Angle between the two regression lines is $\theta = \tan^{-1}\left(\frac{m_1 - m_2}{1 + m_1 m_2}\right)$ where m_1 and m_2 are the slopes of regression lines X on Y and Y on X respectively.
4. The angle between the regression lines indicates the degree of dependence between the variable.
5. Regression equations intersect at $(\bar{X}, \bar{Y})$

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